



Mode I interlaminar fracture toughness of two-dimensional continuous fiber reinforced ceramic matrix composites using wedge-loaded double cantilever beam method

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ABSTRACT

In this study, the wedge-loaded double cantilever beam (W-DCB) method was applied to evaluate the mode I interlaminar fracture toughness of two-dimensional (2D) continuous fiber-reinforced ceramic matrix composites (CMCs). Based on the elastic foundation theory, a W-DCB model was built and its compliance formula was derived by considering the axial force effect, shear effect, asymmetric propagation, and incomplete interlaminar contact. The friction effect was analyzed and measured using the theory of error transfer and a quasi-in-situ measurement method, respectively. The results reveal that rational selection of the wedge angle and friction coefficient can reduce the friction effect. Meanwhile, the W-DCB formula was comprehensively verified by compliance curves, crack tip behaviors, and the cohesive zone of 2D CMCs, which demonstrates its accuracy in conjunction with the experimental data. Finally, the energy release rate (ERR) of the 2D CMCs calculated using the W-DCB theory was consistent with the measured data.

1. Introduction

Continuous fiber-reinforced ceramic matrix composites (CMCs) are important materials for aircraft applications because of their low density, high thermal stability, and excellent mechanical properties. Two-dimensional (2D) CMCs are reinforced by laminated unidirectional strips and plain satin cloth with superior in-plane properties. However, negligible reinforcing fibers between the layers lead to weak interlaminar strength. Consequently, mode I interlaminar fractures are vulnerable to occur under through-thickness loading, low-speed impact, high thermal gradient stress, and geometric discontinuity. Interlaminar fracture toughness (IFT) and the energy release rate (ERR) of mode I are used to characterize the interlaminar crack resistance of CMCs reinforced with 2D preforms, which sometimes were stitched by few fiber yarns, using the double cantilever beam method (DCB) [1–9].

As a suitable candidate for mode I IFT, a wedge-loaded double cantilever beam (W-DCB) method has been applied to the measurement of mode I IFT for PMCs [10–18] and CMCs [19–24], with the advantage of symmetrical opening and low cost of specimen preparation relative to DCB [25]. However, the theoretical model and experimental analysis of

the W-DCB method are still based on the DCB methods for polymer matrix composites (PMCs). Because of the higher porosity and difficulty of machining an idealized pre-crack or notch for CMCs, the previous relevant interlaminar models used in the PMCs should be modified. Importantly, the axial force applied by the wedge indenter plays an important role in the test result of the W-DCB method. Meanwhile, the dynamic friction coefficient between the wedge indenter and test material is always affected by the test environment and method, resulting in significant measurement uncertainty.

To characterize the mode I IFT of 2D CMCs, a mechanical model of the W-DCB method of 2D-CMCs was established according to the elastic foundation beam theory, with considering the material characteristics and loading. Meanwhile, an asymmetric propagation formula was developed to overcome the situation in which cracks deviate from the central plane of the notch because of the difficulty in preparing an ideal pre-crack. The mode I fracture toughness of a 2D carbon-fiber-reinforced silicon carbide composite (2D C/SiC) was measured according to the W-DCB model to validate the model.

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2. Mechanical model for the W-DCB method

2.1. Assumptions of mechanical model

Fig. 1(a) shows a schematic of the W-DCB method. One end of the test specimen was fixed, where the notch end of the test specimen was loaded by a wedge indenter. Consequently, during the translational movement of the wedge indenter along the notch direction, friction must occur at the contact position (approximate line contact) between the wedge indenter and the specimen, as shown in Fig. 1(b). The loading mode is quasi-static, and the balanced equation is given by:

$$\sum X: -2 \times \sin\left(\varphi + \frac{\alpha}{2}\right) \times F_f + F = 0 \quad (1)$$

And,

$$X: \sin\left(\varphi + \frac{\alpha}{2}\right) \times F_f = F_x \quad (2-a)$$

$$Y: \cos\left(\varphi + \frac{\alpha}{2}\right) \times F_f = F_y \quad (2-b)$$

where φ and α are the angles of friction and wedge-indenter, respectively; f and, F_f are friction and applied force between the wedge-indenter and test specimen, respectively; and F , F_x , and F_y are the loading, axial, and opening force, respectively. The force diagram is shown in Fig. 1(c).

Substituting Eq. (1) into Eq. (2), Eq. (3) is derived.

$$F_x = \frac{F}{2} \quad (3-a)$$

$$F_y = \frac{F}{2} \cot\left(\varphi + \frac{\alpha}{2}\right) = \frac{F \cos \frac{\alpha}{2} - \mu \sin \frac{\alpha}{2}}{2 \sin \frac{\alpha}{2} + \mu \cos \frac{\alpha}{2}} \quad (3-b)$$

where the friction coefficient $\mu = \tan \varphi$.

According to Kanninen's opinions on DCB test [26–29], the W-DCB test can also be simplified as a cantilever beam model acting on an elastic foundation, as shown in Fig. 1(d). Compared to the Euler or Timoshenko beam theory, the model based on the cohesive zone or elastic foundation beam theories is more rational and better reflects the delamination failure characteristics of laminates. To facilitate the analysis of the W-DCB model, the following assumptions are introduced:

- 1) The interlaminar bonding is strong enough not to delaminate in the bending process, and the laminates are treated as the orthotropic material.

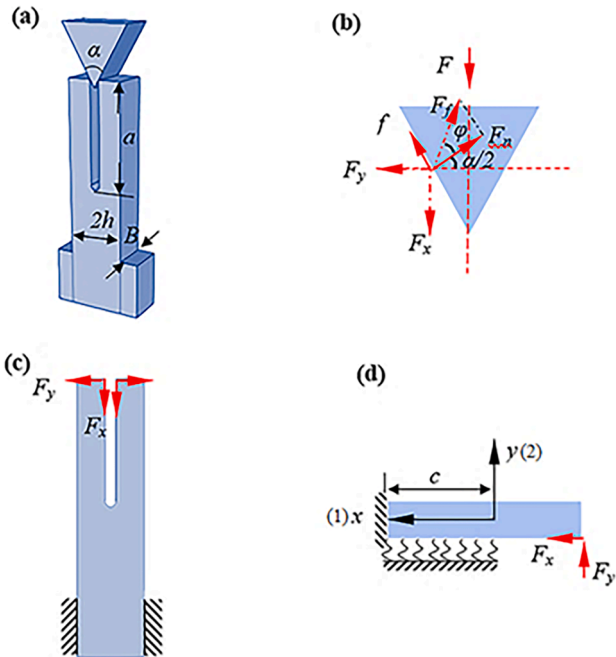


Fig. 1. Mechanical analysis of W-DCB: (a) Mechanical model of W-DCB. (b) Force diagram of wedge-loading. (c) Force diagram of test specimen. (d) Mechanical model of semi-arm of test specimen.

- 2) In the loading process, the only effect of the axial force F_x on the semi-arm beam is the bending moment, and its magnitude is $F_x h/2$. Appendix A provides a detailed analysis.
- 3) The effect of shear force on the mechanical model is considered by a modified coefficient ζ . The analysis is given in Appendix B.

2.2. Solution of the W-DCB mechanical model

Based on the assumptions presented in Section 2.1, the constitutive deformation equation of the W-DCB beam can be expressed as follows.

when $0 \geq x \geq -a$, $w_1(0) = w_2(0)$; $\theta_1(0) = \theta_2(0)$

$$E_{11}I_z \frac{d^2 w_1}{dx^2} = F_y(a+x) - F_x \frac{h}{2} \quad (4)$$

when $c \geq x \geq 0$, $w_2(c) = 0$; $\theta_2(c) = 0$; $w_2'(0) = \frac{F_y a - F_x \frac{h}{2}}{E_{11}I_z}$; $w_2''(0) = \frac{F_y}{E_{11}I_z}$;

$$E_{11}I_z \frac{d^4 w_2}{dx^4} + f(w_2) = 0$$

Considering that the matrix of 2D CMCs is brittle, the elastic-brittle model kw_2 is adopted for the traction-separation constitutive relationship $f(w_2)$ of interlayers, so.

$$E_{11}I_z \frac{d^4 w_2}{dx^4} + kw_2 = 0, \text{ or}$$

$$\frac{d^4 w_2}{dx^4} + 4\lambda^4 w_2 = 0 \quad (5)$$

Herein, w_1 , w_2 are the deflection of the W-DCB beam when $0 > x > -a$ and $c > x > 0$ respectively; θ_1 , θ_2 are the warping angle of semi-arm beam section when $0 > x > -a$ and $c > x > 0$ respectively; a is the crack length including the pre-crack and propagated crack; c is the length along the crack front to the fixed end; $\lambda^4 = k/(4E_{11}I_z)$, where E_{ii} is the elastic modulus of the semi-arm of test specimen in i direction ($i = 1, 2, 3$); I_z is the z -axis moment of inertia, $Bh^3/12$; B is the width of test specimen, h is thickness of semi-arm of W-DCB specimen; k is the stiffness constant of the planar elastic foundation model, which is derived by equivalence principle of force:

$$1 \times kw_2 = \beta \times BE_{22} \frac{w_2}{h/2} \Rightarrow k = \beta \frac{2E_{22}B}{h} \quad (6)$$

where β is a corrected term of planar continuous model in terms of incomplete contact among matrix-foundation and upper (lower) semi-arm beam for the consideration of higher porosity of CMCs, and 1.0 represents a fully contacted state. It's noted that the value of β relies on the braided structure and deposition process of matrix.

By solving Eq. (4), (5) and (6), the deflection curve equations will be.

$$w_1 = F_y \frac{x^3 + 3ax^2}{6E_{11}I_z} - F_x \frac{x^2 h}{4E_{11}I_z} + A_5 x + A_6 \quad (7-a)$$

$$w_2 = A_1 \sin(\lambda x) \sinh(\lambda x) + A_2 \sin(\lambda x) \cosh(\lambda x) + A_3 \cos(\lambda x) \cosh(\lambda x) + A_4 \cos(\lambda x) \sinh(\lambda x) \quad (7-b)$$

The W-DCB model solution is presented in Appendix C. The simplified compliance theory is as follows:

$$\left(\frac{\delta}{F_y}\right)^{\frac{1}{3}} = C^{\frac{1}{3}} = \left(\frac{8}{E_{11}Bh^3}\right)^{\frac{1}{3}} (a + \chi h) = k_s (a + \chi h) \quad (8)$$

where δ is the crack-mouth opening displacement (COD); C is the compliance; k_s is the slope of $C^{1/3}$ - a .

$$\chi = 0.64 \left(\frac{E_{11}}{\beta E_{22}}\right)^{\frac{1}{3}} - 0.25\zeta \quad (9)$$

is called the end correction coefficient relative to built-in model.

Considering the discussion about the calculation of Energy Release Rate, ERR in Appendix A, ERR of W-DCB is

$$G_{I,T} = \frac{\left(\frac{F_y^c}{2B}\right)^2 \frac{dC}{da} = \frac{\left(F_y^c a\right)^2 \left(1 + \chi \frac{h}{a}\right)^2}{E_{11} B I_z} \quad (10)$$

Alternatively, by using the F_y - δ relationship in Eq. (8), we obtain other expression for ERR:

$$G_{I,T} = \frac{3F_y \delta}{2B(a + \chi h)} \quad (11)$$

2.3. Influencing parameters

2.3.1. Friction effect

Mansour et al. [22] studied the friction effect on the result of IFT of 2D CMCs, and demonstrated that selection of the μ and α primarily lies in the ‘advantage of opening force’, or reducing the hindering force of the wedge indenter to move down. Fig. 2(a) shows the relationship between $F/2$ (F_y) and F_y according to Eq. (3-b). According to Mansour’s result, F_y is equal to F through the rational calculation of α and μ ; that is, the ratio of both falls on the 0.5 isoline in Fig. 2(a) with a smaller μ . In fact, the maximum order of magnitude of the mode I interlaminar opening force of CMCs is approximately 100 N, and the total length the wedge indenter moves down does not exceed 10 mm, which is much less than the disc friction used in the reference [22]. Based on the experimental data, μ fluctuates significantly in the process of short-range contact; that is, its value has great uncertainty. Hence, the friction effect lies in the significant uncertainty of the measurement, rather than a huge force that hinders the wedge indenter from moving down.

To reveal the influence of the larger uncertainty of the friction effect on the test results, the transfer theory of errors was introduced. Hence, by deriving Eq. (3-b) with respect to μ , the first order error transfer coefficient (ETC) can be expressed as:

$$\left| \frac{\delta F_y}{F_y} \right| = \frac{\mu}{(\sin \frac{\alpha}{2} + \mu \cos \frac{\alpha}{2})(\cos \frac{\alpha}{2} - \mu \sin \frac{\alpha}{2})} \left| \frac{\delta \mu}{\mu} \right| = ETC \left| \frac{\delta \mu}{\mu} \right| \quad (12)$$

Fig. 2(b) shows the relationship of the ETC with μ and α according to Eq. (12). It can be seen that the value of ETC is mostly <1, and a smoother wedge indenter corresponds to a smaller ETC. Although the relative error of μ is large, the relative error of F_y can be reduced by rational selection of α and μ . This result is consistent with that of Mansour et al.; that is, a wedge indenter with small μ should be selected. Furthermore, the results indicate that μ should be measured at the initial stage rather than at the stage with long-range stability.

In order to measure the friction coefficient between the sample and the wedge indenter, both direct [22] and indirect methods [15] can be selected. The disc method [22] is a common method, but it can not directly and conveniently simulate the in-situ loading mode between the sample and the wedge indenter. Although the indirect method can

reflect the situation of in situ contact, it needs an accurate conversion formula to solve, which has certain errors. Therefore, a method that can easily realize in-situ measurement was invented and applied to measure the friction coefficient of such tests. Fig. 3(a) schematically depicts the measurement method of μ between the specimen and wedge-indenter. The objective of the measurement method is to realize the quasi-in-situ measurement of μ , such as line contact, different contact inclinations, and pressure. The specific measurement process is as follows: One end of the specimen was obliquely placed into a sleeve, and the other end was in line contact with the wedge indenter. Subsequently, a specific value of pressure was applied on the sleeve through the testing machine, and the wedge indenter was slowly driven through a thin line horizontally bypassing the smooth bearing. When the wedge indenter moves uniformly and slowly on the supporting block, μ is given by the balance condition of the wedge indenter:

$$\mu = \frac{f_1}{N} = \frac{T - f_2}{N} \quad (13)$$

where T is tension force of the driving wedge indenter measured from a microsensor depicted by Fig. 3(b); N is the normal force exerted by the specimen on the wedge indenter, which can be controlled and read from the testing machine; f_2 is friction between wedge-indenter and supporting block, which can be measured by replacing the pressure N with different standard weights depicted by Fig. 3(c).

2.3.2. Asymmetric crack propagation

The crack front is blunt due to the mechanical fabrication of notch in CMCs, which leads to ‘crack climbing’ the phenomenon and increases the difference in the thickness of the semi-arm beam on both sides of the crack combined with the notch machining error. If the thickness of a semi-arm beam is h_1 and another is h_2 , according to the Eq. (8), Eq. (14) were obtained.

$$\frac{\delta_1}{F_y} = \left(\frac{4}{E_{11} B h_1^3} \right) (a + \chi h_1)^3 \quad \frac{\delta_2}{F_y} = \left(\frac{4}{E_{11} B h_2^3} \right) (a + \chi h_2)^3 \quad (14)$$

The COD is the sum of δ_1 and δ_2 . Because χh is sufficiently smaller than a , Eq. (14) can be rewritten as:

$$\frac{\delta_1 + \delta_2}{F_y} = K \left(\frac{8}{E_{11} B h^3} \right) a \left(1 + 3\chi \frac{h}{a} \right) = \frac{8K}{E_{11} B h^3} \left(a + \chi h \right)^3 \quad (15)$$

$$\text{in which, } K = \frac{\xi_1 + \xi_2}{2}; \tilde{h} = \frac{\xi_1 h_1 + \xi_2 h_2}{\xi_1 + \xi_2}; \xi_1 = \left(\frac{h}{h_1} \right)^3; \xi_2 = \left(\frac{h}{h_2} \right)^3$$

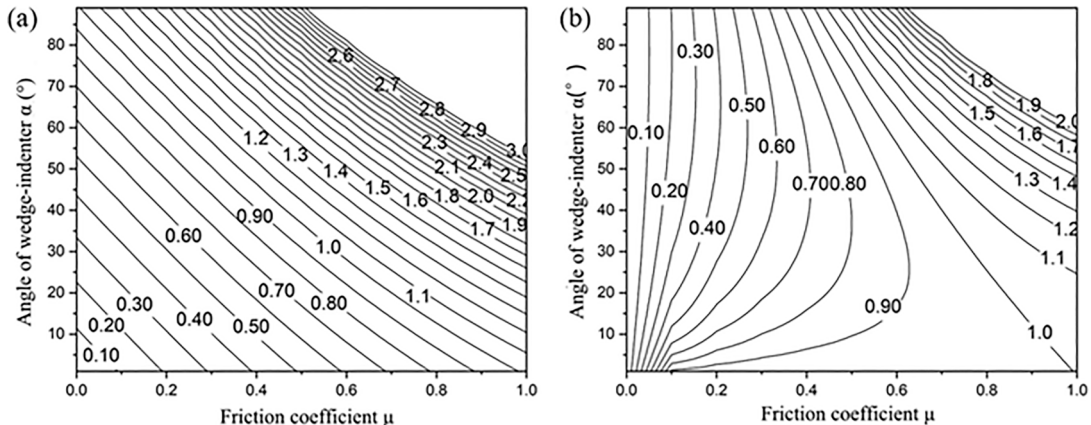


Fig. 2. Friction effect. (a) Ratio of $0.5F$ (F_y) and F_y with μ and α . (b) Relationship of ETC with μ and α .

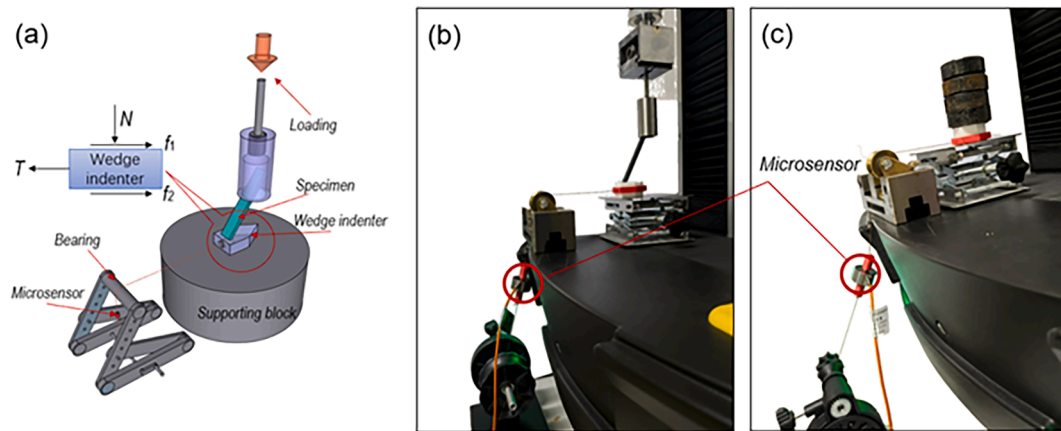


Fig. 3. Test method of μ between test specimen and wedge indenter. (a) Test schematic diagram. (b) Measurement of tension T . (c) Measurement of friction f_2 .

3. Experiments

3.1. Materials and specimen preparation

2D-C/SiC was used to measure the mode I fracture toughness using by the W-DCB method. The composite was reinforced with a T300 plain woven PAN-based carbon fiber. A pyrolytic carbon interface layer with a thickness of approximately 200 nm was prepared by chemical vapor infiltration (CVI), and a SiC matrix was deposited by CVI at 1000 °C. The density, porosity, and fiber volume fraction of the 2D-C/SiC are 2.0 g/cm³, 13 %, and 40 %, respectively.

The specimens used for the mode I fracture toughness were 60 mm long, 10 mm B , and 3 mm h . A short notch as a pre-crack was cut in the middle of the specimen width with a diamond-powder-coated hacksaw in 0.25 mm thick. According to the analysis of the friction effect in Section 2.3.1, zirconia was selected as the material of the wedge indenter because its μ with C/SiC is 0.21 measured by Eq. (13), and it has excellent wear resistance. According to Fig. 2(b), the minimum value of the ETC is approximately 0.4 when α is 60°, and η is calculated to be 0.9 by dividing Eq. (3-a) by Eq. (3-b). E_{11} and E_{22} were measured to be 74,500 and 21300 MPa, respectively, and E_{11}/E_{12} was estimated to be 7–8 based on the result of the numerical simulation [30]. Consequently, the initial crack length a_0 is processed to 11 mm including 1 mm of the additional propagated length, that is, a_0/h is approximately 7–8 based on the analysis in Appendix A and B. Relevant data are listed in Table 1.

3.2. Test procedure

To overcome the machining blunting of the pre-cracks of CMCs, a short (e.g., 1 mm) crack was split in advance at the beginning of the test. During the test, loading and unloading were performed where the crack propagated every 5 mm and the loading force vs displacement data were obtained. The loading speed was 0.5 mm/min. Finally, the critical ERR was calculated using the following formula related to Eq. (11):

$$G_{le} = \frac{3F_y^* \delta}{2B(a + \Delta)} \quad (16)$$

where, F_y^* is the opening force at the point of deviation from the linearity; $\delta \approx 2 \tan(\alpha/2) \times dis$, where dis is the displacement by which the wedge indenter moves down, and Δ is the intercept between the

compliance calibration line and the horizontal axis.

4. Results and discussion

4.1. Interlaminar fracture

Fig. 4(a) shows the fracture morphology of 2D-C/SiC under the wedge-loading. The fracture surface is shown in Fig. 4(b). No obvious fiber bridging was observed during the fracture process. Even if a small part of the fiber breaks, it is brittle fracture and creates short and broken fibers without forming a continuous fiber pull-out phenomenon. Meanwhile, the debonding area occurs near the peak of the braided structure as shown in Fig. 4(b). The reason is that the sinusoidal braided structure of 2D plain weave C/SiC is easy to form a good matrix deposition in the peak area, and a cavity in the valley area. Consequently, real contact area is approximately half of the entire section along the crack or theoretically $\beta = 0.5$. In the crack propagation process, the crack is more likely to propagate along the interface of fibers and matrix owing to the stress concentration effect of deposition cavity. The behavior of the crack passing through the interlayers is schematically plotted in Fig. 4 (c). Owing to the larger machining width of the notch (approximately 10 times the thickness of the single-laminate), the crack has a small probability of propagating along the notch axis. This causes the thicknesses of the upper and lower semi-arm beams to differ on the fracture surface.

4.2. Deflection curve of semi-arm beam

Fig. 5(a) shows the relationship between the F_y -COD curve for different crack lengths. Each curve indicates typical brittle fracture characteristics, that is, the peak point coincides with the knee point from linear to nonlinear, and the peak value of the curves decreases with a . Simultaneously, based on the analysis of the fracture surface in Section 4.1, it is considered that the crack propagation is accompanied by debonding between the matrix and fibers along the fracture section because there is no obvious fiber-bridging, which is in line with linear elastic fracture Mechanics theory (LEFM). Using the parameters in Table 1, the calculated value of COD/ F_y was consistent with the experimental data in Fig. 5(b). Meanwhile, compared to the value calculated by $F_y a^3 / 3E_{11}I_z$ of the fixed end cantilever beam theory, the result shows that its difference with the experimental data increases

Table 1

Experiment parameters of 2D-CVI C/SiC.

E_{11} /MPa	E_{22} /MPa	E_{33} /MPa	a_0 /mm	B /mm	h /mm	h_1 /mm	h_2 /mm	β	α	μ	η
74,500	21,300	74,500	11	10	1.5	1.3	1.7	0.5	60°	0.21	0.9

Note: a-Ultrasonic technique.

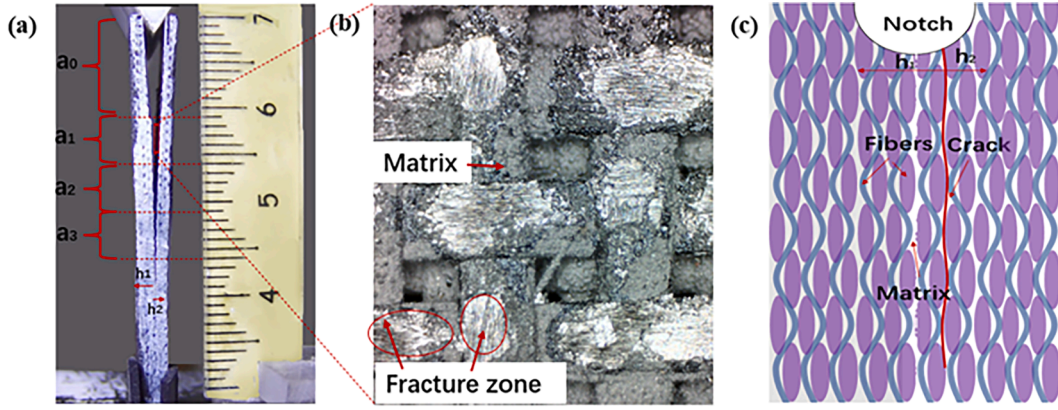


Fig. 4. Interlaminar fractural characteristics of 2D plain weave CVI-C/SiC. (a) Interlaminar fracture of 2D plain weave CVI-C/SiC. (b) Fractural section along the crack route. (c) A sketch map of crack propagation.

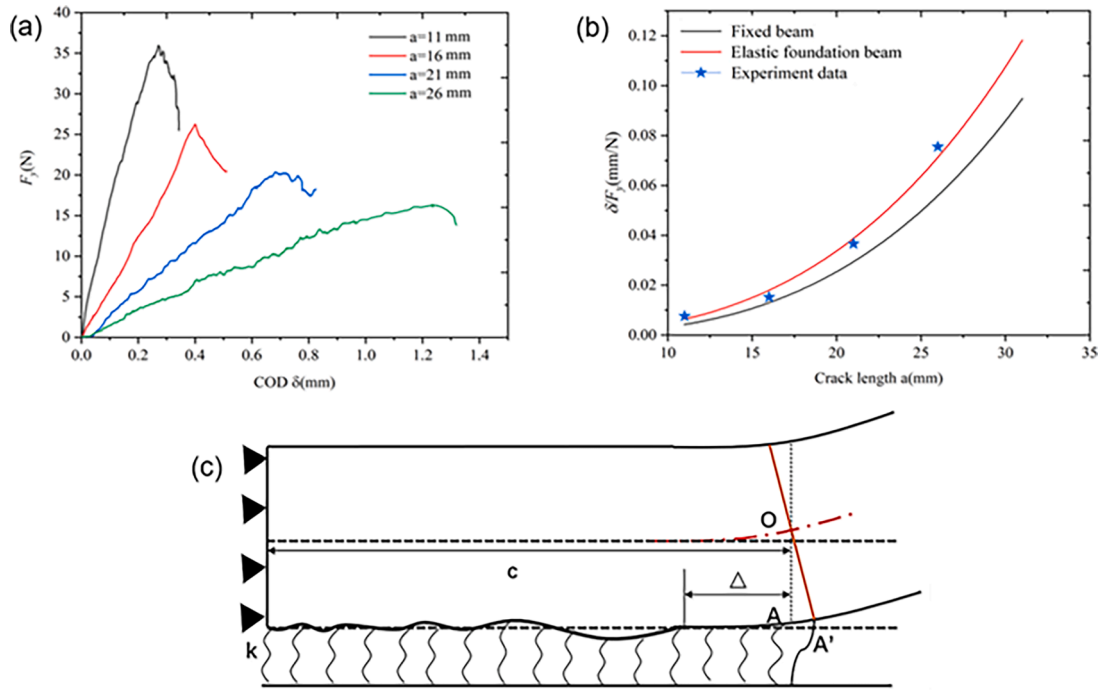


Fig. 5. Compliance curve of W-DCB. (a) F_y -COD curve for different crack lengths. (b) Compliance-crack length. (c) The warping of cross section of the semi-arm at the crack-tip.

with a , while the W-DCB model is consistent with the experimental data. The reason for this result was that the cross section of the semi-arm beam at the crack tip is warped rather than fixed, which proves the rationality of the elastic foundation beam model. It is shown in Fig. 5(c).

4.3. Characteristic of crack tip

Fig. 6 shows the effects of the dimension parameters B and h on the deflection curve of the semi-arm of the W-DCB specimen. When $-11 \text{ mm} < x < 0 \text{ mm}$, the influence on deflection is negatively correlated with the flexural section coefficient I_z ($Bh^3/12$), which is in accordance with the beam bending theory. For $0 \text{ mm} < x < 35 \text{ mm}$, that is, the elastic foundation zone, the relationship between the laminates along the crack alternate between traction and separation, which is also shown in Fig. 5(c). The first traction zone was called as the cohesive zone (CZ) and its length is treated as the cohesive zone length. In the CZ, its length is positively correlated with h , as verified by the CZM theory [16,31]. Because its magnitude is only micron level in Fig. 6, this explains why

the crack tail can be observed using the DIC technique [6,32]. Meanwhile, based on the analysis that the behavior of the crack -tip satisfied the LEFM theory in Section 4.2, the maximum tension stress $T_{\max}$ (MPa) in the CZ is, neglecting the third-order small quantity of (h/a) in Eq. (C.3) about $w(0)$:

$$T_{\max} = 2 \frac{\beta E_{22}}{\tilde{h}} w_2(0) = 2 \frac{\beta E_{22}}{\tilde{h}} \times \frac{F_y a^3}{3 E_{11} I_z} 0.61 \left(\frac{\tilde{h}}{a} \right)^2 \sqrt{\frac{E_{11}}{\beta E_{22}}} = 4.88 \sqrt{\frac{\beta E_{22}}{E_{11}}} \frac{F_y a}{B \tilde{h}} \quad (17)$$

When $T_{\max}$ is equal to the transverse tensile strength (TTS), the crack propagates. Through loading, the corresponding crack -onset force at different crack lengths can be obtained, and TTS is calculated by Eq. (17), as listed in Table. 2.

According to the above result, the TTS 36.5 MPa is almost three times than the through-thickness tensile strength 13.4 MPa for 2D C/SiC [33]. It is difficult to quantitatively evaluate the causes of the errors, because of the small disturbance of the parameters in Eq. (17), especially the

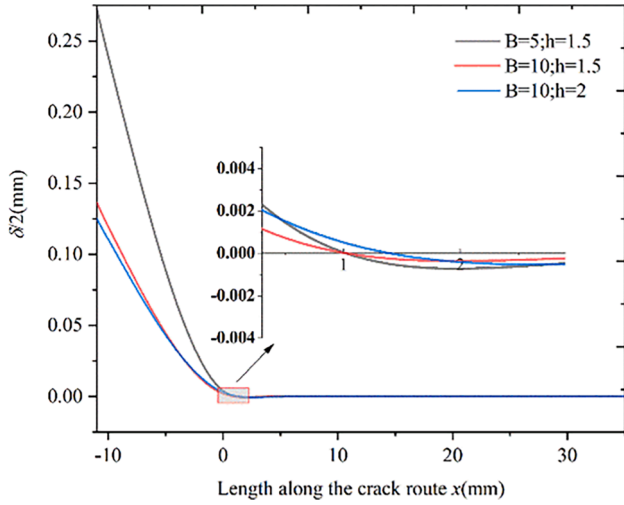


Fig. 6. Deflection changes of semi-arm beam at different zones.

Table 2

The calculation of TTS.

F_y^c/N	a/mm	$w(0)(\mu m)$	$T_{max(av.)}/Mpa$
36.0	11.0	1.80	36.5
26.2	16.0	1.84	
20.4	21.0	1.85	
16.4	26.0	1.79	

parameter h , has a significant impact on the results. Importantly, the TTS is a micro description, whereas the through-thickness tensile strength is a macro illustration. Simultaneously, there were statistically significant differences between them. However, the magnitudes of the TTS can be estimated from Table 2, and the results also reflect the consistency of the 2D CMCs in the process of interlaminar fracture.

4.4. Compliance calibration and ERR

Fig. 7(a) illustrates the compliance calibration curve, which indicates an excellent linear distribution. The model curve is in excellent agreement with the experimental data obtained by substituting the data in Table 1 into Eq. (15). The result indicates that the end correction term obtained by data fitting is $\Delta=1.22$ mm and the theoretical prediction result is 1.16 mm. Meanwhile, practically, the modulus E_{22} is difficult to measure even if the ultrasonic technique can be used. Making use of Eq. (9), E_{22} can be calculated as follows:

$$E_{22} = \frac{1}{\beta} \left(\frac{\Delta + 0.25\zeta}{h} \right)^{-4} E_{11} \quad (18)$$

The ERR can also be calculated according to the compliance calibration curve derived using Eq. (10):

$$G_{I,Com} = \frac{(F_y^c)^2}{2B} \frac{dC}{da} = \frac{3k_s}{2B} (F_y^c)^2 C^3 \quad (19)$$

Simultaneously, the modulus E_{11} estimated using Eq. (8) and Eq. (15) is given by

$$E_{11} = \frac{8K}{Bh^3k_s} \quad (20)$$

Consequently, using Eq. (20), the Eq. (18) can be rewritten as

$$E_{22} = \frac{1}{\beta} \left(\frac{\Delta + 0.25\zeta}{h} \right)^{-4} \frac{8K}{Bh^3k_s} \quad (21)$$

And considering the asymmetric propagation, theoretical calculation formula Eq. (10) is rewritten as:

$$G_{I,T} = \frac{(F_y^c)^2}{2B} \frac{dC}{da} = \frac{K(F_y^c)^2(a + \chi\tilde{h})^2}{E_{11}Bl_c} = \left(\frac{F_y^c a}{Bh^2} \right)^2 \frac{12K(1 + \chi\tilde{h}/a)^2 h}{E_{11}} \quad (22)$$

Fig. 7(b) shows the change in ERR with the crack length calculated using Eqs. (16), (19) and (22). The theoretical results are consistent with the experimental data; however, there are still some differences owing to the inconsistency of TTS and contact area between interlayers. In contrast, the theoretical results demonstrated excellent consistency owing to the use of unchanged constants. Even so, the theoretical calculation still better reflects the actual value. The ERR is closely related to TTS from the formal point of view of Eqs. (17), (22). Through a simple derivation, a formula similar to CZM theory can also be obtained. To verify the TTS estimated using Eq. (17), the CZ length L_{cz} is estimated by CZ theory [16] and ERR, that is, the end correction term, $L_{cz} =$

$0.84 \left(\frac{E_{22}G_{IC}}{T_{max}^2} \right)^{0.25 \sim 0.75} h = 1.25mm$. The results also confirm the W-DCB model.

5. Conclusions

The wedge indenter acts directly on the notch of the DCB sample to split the sample, which is an important alternative method to the traditional DCB method. However, owing to the introduction of μ and F_y , the W-DCB method is bound to have different influencing parameters

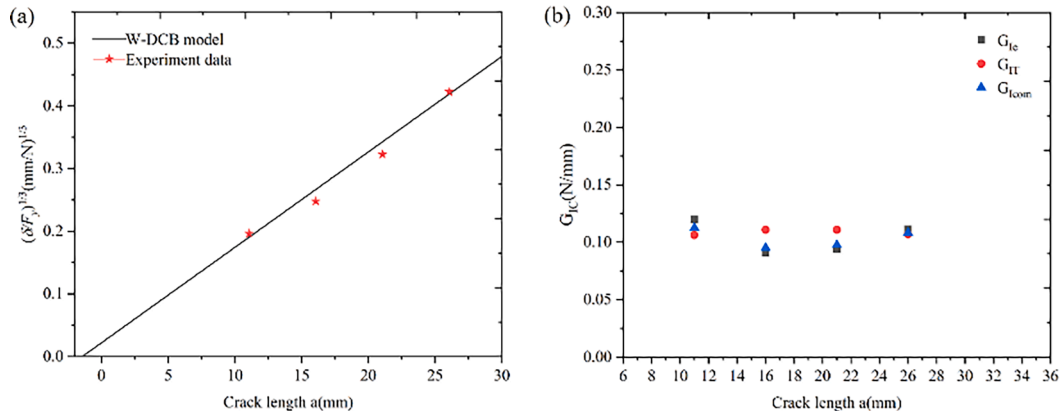


Fig. 7. IFT of 2D plain weave C/SiC. (a) Compliance calibration. (b) ERR(0.107 ± 0.017 N/mm).

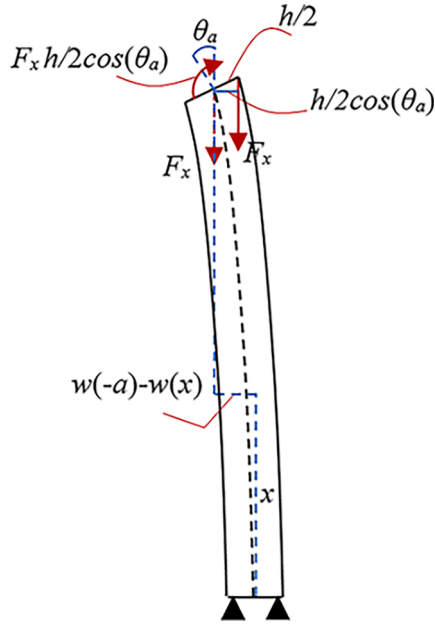


Fig. A1. Axial force effect of W-DCB.

and a theoretical analysis than the DCB method. A mechanical model, experimental test and result analysis were introduced to evaluate the W-DCB mechanical model and method in this study. The following conclusions were drawn from this study:

- The mechanical model of the W-DCB was built using elastic foundation theory to obtain expressions for the deflection and warping angle at the loading end and crack front, respectively. The results revealed that axial force F_x modified the end correction term of W-DCB relative to the compliance formula to the DCB. Simultaneously, based on the LEFM theory, the TTS can be evaluated using the W-DCB model.
- An in-situ test method was proposed to measure μ . Simultaneously, the error transfer theory was used to evaluate the influence of a larger uncertainty of μ . The results indicated that rational selection of the wedge angle can reduce this adverse effect.
- The proposed W-DCB test method was used to perform the interlaminar splitting test of the 2D plain weave CVI-C/SiC material. The results demonstrated that the cross section of the tested specimen was incomplete depending on the braided structure and deposition process, and there was no obvious fiber bridging but instead brittle fracture in the damage zone. The

Appendix A. Axial force effect

Because the specimen was loaded by the wedge-shaped indenter, the specimen was not only subjected to the opening force F_y perpendicular to the crack direction, but also the axial force F_x parallel to the crack direction. According to the mechanical model established in Section 2.1, the F_x can be simplified to a concentrated force F_x and a couple $F_x \times h/2 \cos(\theta_a)$ towards the central axis of the semi-arm beam, in which θ_a is warping angle of cross section when $x = -a$. The effect of F_x is schematically shown in Fig. A1. According to theory of elasticity and superposition principle about fixed end cantilever beam [34], the warping angle of cross section of beam at the point $(-a, 0)$ is

$$\theta_a = -\frac{F_y a^2}{2E_{11}I_z} \left[1 - \eta \frac{h}{a} + 0.25 \left(\frac{E_{11}}{E_{12}} \frac{h}{a} \right) \frac{h}{a} \right] \quad (\text{A.1})$$

where each item in the square brackets from left to right is the warping angle caused by F_y , F_x , and the shear force, respectively; $\eta = F_x/F_y$.

Apparently, when a/h is 10 and $E_{11}/E_{12} = 10$, the warping angle of beam section caused by axial effect is the same level as that caused by shear force, and symbols of both are opposite. Meanwhile, the contribution of F_x on θ_a is some times than the shear force by evaluating the Eq. (A.1). Hence, a modified coefficient ζ_1 is introduced to simplify Eq. (A.1) as follows:

reason is that the fibers in the plain weave structure are local short fibers rather than long fibers, which affects the mechanical properties of the ceramics.

- Compared to the experimental data, the results indicated that the W-DCB mechanical model is consistent with test results, including the deflection curve and energy release rate. Meanwhile, the formula of interlaminar TTS was verified by CZM theory in conjunction with the experimental data.

Author Contribution roles

Wei-gang SHI: Mechanical model building and solutions; Original draft.

Chao ZHANG: Funding; Experiment and data analysis.

Bo WANG: Providing simulation results relevant to the calculated parameters.

Mei LI: Providing the test specimens and discussion about the results.

Cheng-yu ZHANG: Funding.

CRediT authorship contribution statement

Weigang Shi: . **Chao Zhang:** . **Bo Wang:** . **Mei Li:** . **Chengyu Zhang:** Writing – review & editing.

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Data availability

No data was used for the research described in the article.

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$$\theta_a = -\frac{F_y a^2}{2E_{11}I_z} \left(1 - \zeta_1 \frac{h}{a}\right) \quad (\text{A.2})$$

where the magnitude of ζ_1 depends on the value of $\eta - 0.25 \frac{E_{11}}{E_{12}} \frac{h}{a}$.

Simultaneously, considering the order of magnitude of F_y , $(h/a)^2$, and E_{11} was $10^1 \sim 10^2$, 10^2 , and $10^4 \sim 10^5$, the order of magnitude of warping angel θ_a was nearly 10^{-1} that $\cos(\theta_a) \approx 1$ using Eq. (A.2). Hence, the bending moment of axial force F_x is rationally $F_x \times h/2$. Besides, the concentrated force F_x has a bending moment $F_x \times [w(-a) - w(x)]$ on the beam section. However, since its maximum value is only $F_x \times w(-a)$ and a is so large, which is the same level as $F_x \times h/2$, it is also ignored.

Importantly, to evaluate the contribution to ERR of axial force is necessary. Using William's analysis method about ERR [35], the definition of ERR is

$$G = \frac{d(-U_e)}{da} \Big|_{P=\text{const}} \quad (\text{A.3})$$

where U_e is the elastic potential energy brought by the bending moment, axial force and shear force totally; Eq. (A.3) denotes that the ERR is the change rate of elastic potential with crack length when external force is constant.

Considering the load condition of the W-DCB model as shown in Fig. 1(d), we can get.

$$G^M = \frac{1}{B} \left[\frac{12M_1^2}{2EBh_1^3} + \frac{12M_2^2}{2EBh_2^3} - \frac{12(M_1+M_2)^2}{2E(2h)^3} \right] = G_I^M + G_{II}^M \quad (\text{A.4-a})$$

$$G_{II}^N = \frac{1}{B} \left[\frac{F_y^2}{2E_{11}Bh_1} + \frac{F_y^2}{2E_{11}Bh_2} - \frac{4F_y^2}{2E_{11}B2h} \right] = Er_{II}^N \frac{1}{12} \eta^2 \left(\frac{h}{a} \right)^2 \frac{(F_y a)^2}{BE_{11}I_z} \quad (\text{A.4-b})$$

$$G_I^Q = \frac{3}{10} \frac{1}{E_{12}B^2h} \left[\frac{1}{t} \left(\frac{dM_1}{da} \right)^2 + \frac{1}{1-t} \left(\frac{dM_2}{da} \right)^2 - \left(\frac{dM_2}{da} + \frac{dM_1}{da} \right)^2 \right] = \frac{(F_y a)^2}{BE_{11}I_z} \left[\frac{1}{10} \frac{E_{11}}{E_{12}} \left(\frac{h}{a} \right)^2 \right] \quad (\text{A.4-c})$$

in which,

$$\begin{aligned} G_I^M &= \frac{1}{BE_{11}I_z} \frac{(M_2 - \psi M_1)^2}{16(1-t)^3(1+\psi)} = \frac{(aF_y)^2}{BE_{11}I_z} \left[\left(Er_{I,1}^M - Er_{I,12}^M \eta \frac{h}{a} \right) + Er_{I,2}^M \eta^2 \left(\frac{h}{2a} \right)^2 \right] \\ G_{II}^M &= \frac{1}{BE_{11}I_z} \frac{3(1-t)(M_1+M_2)^2}{16t^2(\psi+1)} = Er_{II}^M \frac{1}{4} \left(\frac{h}{a} \right)^2 \eta^2 \frac{(F_y a)^2}{BE_{11}I_z} \\ Er_{I,1}^M &= \frac{(3\xi^2+1)}{(1-\xi^2)^3}; Er_{I,12}^M = \frac{(1+\xi^2)}{(1-\xi^2)^2}; Er_{I,2}^M = \frac{(1+\xi^2)^2}{(1-\xi^2)(3\xi^2+1)}; \\ Er_{II}^M &= \frac{3(1-\xi^2)\xi^2}{2[(1+\xi)^3 + (1-\xi)^3]}; Er_{II}^N = \frac{\xi^2}{1-\xi^2} \end{aligned} \quad (\text{A.5})$$

where the subscript I, II represent the mode I and mode II fracture, respectively; subscript M, N, Q represent the moment, axial force, and shear force, respectively; $t = h_1/2h$; $M_1 = (M_2 - \psi M_1)/(1 + \psi)$; $M_{II} = (M_1 + M_2)/(1 + \psi)$; $\psi = h_1^3/h_2^3$; $\xi = d_{\text{off}}/h$, and d_{off} is offset distance from the center line. $M_1 = F_y a - F_x h_1/2$; $M_2 = -(F_y a - F_x h_2/2)$; $M = M_1 + M_2$.

Through calculation of Eq. (A.4) and Eq. (A.5), mode II fracture was not existing compared Er_{II}^M with $Er_{I,1}^M$; and the contribution to ERR of axial force was negligible because Er_{II}^N was so smaller than $Er_{I,1}^M$. However, the value of $Er_{I,12}^M$ was same level than $Er_{I,1}^M$, so the axial force effect on the G_I^M was not negligible as a corrected term in the ERR calculated by opening force F_y .

Appendix B. Shear force effect

Based on the theory of elasticity and superposition principle about fixed end cantilever beam [34], the deflection of beam at the point $(-a, 0)$ is

$$w(-a) = \frac{F_y a^3}{3E_{11}I_z} \left[1 - 0.75\eta \frac{h}{a} + 0.38 \left(\frac{E_{11}}{E_{12}} \frac{h}{a} \right) \frac{h}{a} \right] \quad (\text{B.1})$$

where each item in square brackets from left to right is the deflection caused by F_y , axial force F_x , and shear force effect, respectively. Based on the analysis same as Appendix A, the modified coefficient is introduced:

$$w(-a) = \frac{F_y a^3}{3E_{11}I_z} \left(1 - 0.75\zeta_2 \frac{h}{a} \right) \quad (\text{B.2})$$

where the magnitude of ζ_2 depends on the value of $\eta - 0.5 \frac{E_{11}}{E_{12}} \frac{h}{a}$.

Appendix C. Mechanical model of W-DCB solutions based on elastic foundation

With boundary conditions of Eqs. 4 and 5, undetermined coefficients $A_1 \sim A_6$ of Eq. (7) can be solved. Moreover, considering the condition $(c/h) > 5$ and $E_{11}/E_{22} = 1$, the minimum of $\exp(2c\lambda) \gg 1$, we can get:

$$\frac{\delta/2}{F_y} = C_0 \left\{ \left[1 + 1.92 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \frac{h}{a} + 1.22 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} \left(\frac{h}{a} \right)^2 + 0.39 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{3}{4}} \left(\frac{h}{a} \right)^3 \right] - \eta \left[0.75 \frac{h}{a} + 0.96 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \left(\frac{h}{a} \right)^2 + 0.31 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} \left(\frac{h}{a} \right)^3 \right] \right\}, \text{ or}$$

$$\frac{\delta/2}{F_y} = C_0 \left[1 + \left(1.92 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - 0.75\eta \right) \frac{h}{a} + \left(1.22 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - 0.96\eta \right) \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \left(\frac{h}{a} \right)^2 + \left(0.39 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - 0.31\eta \right) \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} \left(\frac{h}{a} \right)^3 \right] \quad (C.1)$$

$$\theta(-a) = -\frac{a^2 F_y}{2E_{11}I_z} \left[1 + \left(1.28 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - \eta \right) \frac{h}{a} + \left(0.64 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - \eta \right) \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \left(\frac{h}{a} \right)^2 \right] \quad (C.2)$$

$$w(0) = \frac{a^3 F_y}{3E_{11}I_z} \left[0.61 \left(\frac{h}{a} \right)^2 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} + 0.39 \left(\frac{h}{a} \right)^3 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{3}{4}} - 0.31\eta \left(\frac{h}{a} \right)^3 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} \right] \quad (C.3)$$

$$\theta(0) = -\frac{a^2 F_y}{2E_{11}I_z} \left[1.28 \frac{h}{a} \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} + 0.82 \left(\frac{h}{a} \right)^2 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} - 0.64\eta \left(\frac{h}{a} \right)^2 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \right] \quad (C.4)$$

where $C_0 = a^3/(3E_{11}I_z)$, θ is warping angel of cross section of semi-beam.

When h is small enough than a , the equation (C.5-C.8) can be further simplified to,

$$\frac{\delta/2}{F_y} = C_0 \left[1 + \left(1.92 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - 0.75\eta \right) \frac{h}{a} \right] \quad (C.5)$$

$$\theta(-a) = -\frac{a^2 F_y}{2E_{11}I_z} \left[1 + \left(1.28 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} - \eta \right) \frac{h}{a} \right] \quad (C.6)$$

$$w(0) = 0.61 \frac{a^3 F_y}{3E_{11}I_z} \left(\frac{h}{a} \right)^2 \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{2}} \quad (C.7)$$

$$\theta(0) = -1.28 \frac{a^2 F_y}{2E_{11}I_z} \frac{h}{a} \left(\frac{E_{11}}{\beta E_{22}} \right)^{\frac{1}{4}} \quad (C.8)$$

Based on the analysis of Appendix A and Appendix B about shear effect by introducing the coefficient ζ , so η in Eq. (C.5-C.6) should be replaced by ζ .

As for the value of ζ , considering that the crack length will gradually increase, it suggests that ζ should take the average of the initial calculation value and the termination calculation value. Taking the compliance calculation formula (C.5) as an example, combined with the analysis of Appendix B, if the initial a/h is equivalent to E_{11}/E_{22} , ζ is η -0.5; If h/a is reduced to half of the initial value at the end of test, ζ is η -0.25. Therefore, ζ is the average of both, that is, η -0.37.

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